Mohammad Ghufran

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EDUCATION

Master of Engineering in Robotics and Automation, GPA: 3.714 **Aug 2022 - May 2024**

University of Arizona

Relevant Coursework: Q-Learning Reinforcement Technique, K-Means Unsupervised Machine Learning, System Identification, Path Planning, Stability & Modern Control

Bachelor of Technology in Mechanical Engineering, GPA: 3.48 **Aug 2018 - May 2022**

Aligarh Muslim University, India

Coursework: Manufacturing Techniques, Computational Fluid Dynamics, Structural Analysis, Computer-aided Design, Operations Research, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, C++, C, HTML, JavaScript, CSS, MATLAB
Software: SOLIDWORKS, Fusion 360, AutoCAD, DS Abaqus, ANSYS, Adobe Creative Cloud, ROS
Tools & Skills: CAD Modelling, Data Analysis, 3D-Printing, Casting, Soldering, Lathes

ACADEMIC EXPERIENCE

University of Arizona **Nov 2022 - Present**

Student Researcher

- Utilize SOLIDWORKS to design and optimize drone designs, ensuring aerodynamics, stability, and functionality through precise modeling and simulation, result include **reducing the size by 50%** without compromising performance
- Deploys the Astar Algorithm in Python for multi-agent systems, achieving optimized functionality, efficient pathfinding, and refined performance in various environments
- Simulate swarms of drones using Gazebo, enabling an accurate representation of their behavior and performance, validating the system, identifying potential issues, and reducing hardware adjustments and expenses

ACADEMIC PROJECTS

Stem Curriculum Recommender using K-Means Algorithm **March 2023 - May 2023**

- Established a project goal to contribute to **Sustainable Development Goal 4 (UN),** Quality Education, by developing a regression model that matches educational content with specific topics in a curriculum
- Developed an unsupervised machine learning model using K-Means Clustering and TF-IDF vectorization techniques, successfully clustering diverse topics from K-12 open-sourced educational resources into distinct categories based on content similarity, thereby obtaining a 95% accuracy in topic clustering

One-Dimensional Ultrasound Guidance via Q-Learning **March 2023 - May 2023**

- Engineered an interactive system comprising a PIC, motor driver, ultrasound sensor, and RS232 connector, assembled on a breadboard
- Leveraged C programming for PIC16F690 and MATLAB for Q-Learning in the development and training of a machine learning model designed to interpret user behavior based on ultrasonic signals

Reconfigurable UAV **Nov 2022 - Present**

University of Arizona

- Utilized SOLIDWORKS to design various components, resulting in the improved structural integrity of the final model, ultimately contributing to performance, efficiency, and product quality
- Conducted in-depth assessments of various autopilot platforms and successfully implemented the most reliable and high-performing solutions, resulting in optimized flight control and reliability of the UAV

Development of Sanitization Drone **Aug 2021 - April 2022**

Final Year Project | Best Project Award

- Utilized experience in resolving complex issues in creative, efficient, and effective ways to create an aerodynamic drone with Solidworks, verified by 3D printing a design model and simulating it with Ansys to achieve a factor of safety of more than 10
- Validated the simulation results in a wind tunnel to ensure the drone's stability and optimal performance

PROFESSIONAL EXPERIENCE

Engineering & Environmental Solutions Pvt Ltd., India **June 2022 - July 2022**

Drone Development Intern

- Developed and led testing of an agriculture drone integrating IFFCO's nano urea, **reducing required fertilizer by up to 60%**, in collaboration with agriculture experts to ensure usability, resulting in reduced costs
- Demonstrated excellent written and oral communication skills in collaborating with cross-functional teams to create a DGCA-compliant user and maintenance manual, which was integral in obtaining product approval and included comprehensive safety, operation, and maintenance information

RESEARCH PAPER

Uppaluru, H., Ghufran, M., El Asslouj, A., & Rastgoftar, H. (2023). "**Drones Practicing Mechanics**" Paper to be presented at the 2023 International Conference on Unmanned Aircraft Systems (ICUAS), Poland